Current Clarity Stage 16: Fresh Reliability Validation

Diagnostic only.

- The frozen engine is unchanged: no refit, every Stage 14/15 file matches its hash, and no model coefficient changed.
- Production is unchanged, and nothing was deployed.
- Excluded from every step:
  - Michael's Guntersville survey and Sector field reports;
  - the Stage 14/15 certification set.
- Water Temperature stays paused.

Freeze order (sha256 values in `FREEZE_STAGE16.json`):

Step	UTC	What was fixed
Pre-registration	2026-10-01T17:26:53Z	Gate method, tier-2 options, selection rule, fresh-test protocol, minima, scores, bars (`PREREGISTRATION_STAGE16.md`, `cf7a8d91…`)
Gate and tier-2 rule	2026-10-01T17:28:55Z	Thresholds and posteriors from development rows only (`engine/rule16.json`, `ed1142fb…`)
Fresh sample	2026-10-01T17:42:14Z	`fresh/fresh_sample.csv` (`02fbe25f…`), built value-blind
ACOLITE results	2026-10-01T17:54:56Z	`fresh/aco_manifest_fresh.txt`
Unsealed	2026-10-01T17:55:32Z	Fresh Secchi, turbidity and chlorophyll (`fresh/truth16.json`, `3b77ffe9…`)

The answer

The reliability gate removes most of the harmful answers. The fresh holdout is too small to certify it.

- **On development, the gate cuts harmful answers.** A state is issued only when its calibrated posterior is at least 70%. That takes harmful answers from 16.6% to **10.6%** and severe ones from 2.0% to 1.5%, with the operational score steady at 7/10.
- **The gate never issues a plain "Normal" or "Stained".** Neither ever reaches 70% confidence, so the evidence removed them, not a hand-written rule.
- **Tier 2 is now boundary states only.** Its harmful rate on development falls from 12% (Stage 15) to 7%. The Stage 15 re-score had shown 29%.
- **Stained water still scores 0.571 on development,** below the 0.60 bar. No threshold choice fixes it, because it is the retrieval-resolution limit.

The fresh holdout is all of the P2 pool that exists today. It is 2026 WQP data at sites no stage has touched, and it yields only **108 observations**:

- 88% of them are from Wisconsin, 53% are Sentinel-2C and 69% are clear water;
- normal (19 of 40), stained (9 of 60), limited (6 of 25) and the total (108 of 200) are all **INSUFFICIENT**.

What the 108 rows show:

- **Score.** 7/10 operational, with one harmful answer (0.9%). The Stage 15 semantics on the same rows had 4.6% harmful.
- **Why harm is so low.** Half the rows are Sentinel-2C at lower authority, and those cannot be harmful by construction.

The verdicts:

- **Exact feet** score 2/10 on the fresh set, mostly because S2C feet are withheld.
- **Sector Intelligence: NO.** Only 5 answers qualify as strong evidence (30 needed).
- **Readiness to freeze is NO** for two reasons:
  - the fresh minima are not met;
  - development stained misses 0.60.

1. The single-state reliability gate (development only)

The posterior model.

- **The fits.** Three monotone logistics give P(truth ≥ 6, 3, 2 ft | prediction).
- **The class posteriors** follow from their differences.
- **Fitting.** Nested in folds for evaluation; on all development lake-held-out predictions for production.

The rule. A single state requires a posterior ≥ τ_1; otherwise the best adjacent pair requires ≥ τ_2; otherwise Unknown.

- **The search.** A pre-registered grid of 128 $\tau_1 \times \tau_2 \times$ tier-2 combinations.
- **No combination met the primary constraints** (harmful ≤ 12%, severe ≤ 1.5%, every class ≥ 0.60), because stained stays below 0.60 everywhere.
- **The fallback rule** (the best score with harmful ≤ 15% and severe ≤ 2%) chose **$\tau_1 = \tau_2 = 0.70$** with tier-2 option **T2-D**.

Development (nested lake-held-out)	Stage 15 semantics	Stage 16 gate
Operational (class-balanced)	0.688	**0.693** (7/10)
Harmful	16.6%	**10.6%**
Severe	2.0%	**1.5%**
Clear	0.848	**0.880**
Normal	0.690	**0.643**
Stained	0.563	**0.571**
Limited	0.652	**0.676**

What the frozen rule issues at tier 1:

Prediction (tier 1)	Issued state
0.50 ft to 1.55 ft	Limited
1.55 ft to 2.43 ft	Stained-to-Limited
2.43 ft to 2.57 ft	Unknown (Stained-to-Limited)
2.57 ft to 2.80 ft	Unknown (Normal-to-Stained)
2.80 ft to 4.11 ft	Normal-to-Stained
4.11 ft to 7.40 ft	Clear-to-Normal
7.40 ft and up	Clear

State issued (development)	n	Right
Clear	531	89%
Clear-to-Normal	397	89%
Normal-to-Stained	215	78%
Unknown / Low Authority	203	–
Stained-to-Limited	179	84%
Limited	57	88%

- **Plain states left.** The only plain states it ever issues are Clear (above 7.40 ft) and Limited (below 1.55 ft).
- **Where it says Unknown.** Between 2.43 and 2.80 ft, where neither boundary pair reaches 70%. That is 93 tier-1 development answers (6%); with tier 3's Unknowns, Unknown is 13% of all answers.

2. Tier-2 authority (development only)

Tier-2 option (at $\tau_1 = \tau_2 = 0.70$, development)	Operational	Overall harmful	Tier-2 harmful
T2-A: Full authority, tier-1 posteriors	0.683	10.7%	8.4%
T2-B: Full authority, tier-2 posteriors	0.692	11.0%	10.5%
T2-C: Lower authority	0.669	9.7%	0.0%
T2-D: Boundary states only (tier-2 posteriors), else Unknown	0.693	10.6%	7.3%

The selected option is boundary-only. Tier-2 answers get posteriors calibrated on tier-2 rows, issue only boundary states, and are Unknown below 70%.

- **The effect.** Tier-2 harmful on development falls from 12% (Stage 15 semantics) to 7%, at the best operational score of the four options.
- **The alternative.** Making tier 2 "lower authority" would remove its harm entirely but lower the score. The pre-registered rule preferred correctness through calibration, not flags.

3. Sentinel-2C (unchanged)

The temporary lower-authority rule stays, with no recalibration. The fresh holdout adds independent S2C evidence:

- **Fresh rows.** 45 S2C rows at tiers 1–2, median signed log **-0.092** against -0.054 for S2A/S2B in the same months.
- **Combined.** With Stage 14's 23 certification rows (–0.19), that makes 68 independent S2C rows.
- **The lift criterion is not met:** ≥ 60 rows with |median signed log| ≤ 0.05. S2C still reads low, by about 0.04 log more than S2A/S2B on fresh water.

4. The fresh holdout

Sources.

- **The pull.** WQP "Depth, Secchi disk depth" at lake, reservoir and impoundment sites, dated 2026-01-01 to 2026-09-28. It was re-fetched on 2026-10-01 with the unchanged Stage 1 downloader and cleaning; only the start date changed.
- **WQP had published nothing new since 27 September:** 4,773 rows both times.

Exclusions.

- Any site in an earlier development or certification file;
- the Guntersville box;

- censored readings.

The lake rule. It is the Stage 10 P2 definition: sites in no earlier file. The strict lake-independent subset is reported apart.

Fresh pool step (value-blind)	Count
WQP Secchi rows dated 2026, fetched 2026-10-01	4,773
… at sites in an earlier development/certification file (excluded)	3,278
Eligible site-days	1,486
One per site-month	1,111 (419 sites, 265 lakes; Wisconsin 926)
Clear pass within ±1 day (≤ 20% cloud)	305
Matched (≥ 5 valid pixels)	126
Fresh sample (≤ 3 per lake)	**108** at 80 lakes
… Wisconsin	95 (88%)
… Sentinel-2C	57 (53%)
… lake-independent (strict)	22

Core class	Fresh n	Minimum (pre-registered)	
Clear	74	40	met
Normal	19	40	**INSUFFICIENT**
Stained	9	60	**INSUFFICIENT**
Limited	6	25	**INSUFFICIENT**
Scored rows	108	200	**INSUFFICIENT**

The verdict is INSUFFICIENT. The minima were fixed before any value was read and are not lowered. Everything below the minima table is reported as information only.

5. What the fresh rows show (information; not verdict-bearing for the short classes)

Fresh holdout (one scoring)	Stage 15 semantics (comparison)	Stage 16 product
Operational (class-balanced)	0.667	**0.709** (7/10)
Harmful	4.6%	**0.9%**
Severe	0.0%	**0.0%**
Clear (n 74)	0.809	**0.826**
Normal (n 19)	0.576	**0.529**
Stained (n 9)	0.506	**0.706**
Limited (n 6)	0.775	**0.775**

Outcome (fresh)	Rows	Share
Correct single state	27	25%
Correct boundary state	16	15%
Correct lower-authority answer	47	44%
Wrong lower-authority answer	8	7%
Appropriate Unknown	5	5%
Unknown that withheld a right answer	4	4%
Harmful wrong answer	1	1%

State issued (fresh)	n	Right
Clear	51	96%
Clear-to-Normal	23	100%
Normal-to-Stained	11	82%
Stained-to-Limited	9	67%
Unknown / Low Authority	9	–
Limited	5	60%

Split (fresh)	Group	n	Operational	Harmful	Classes
platform	sentinel-2b	43	0.691	2.3%	clear 32, limited 3, normal 5, stained 3
platform	sentinel-2c	57	0.657	0.0%	clear 38, normal 14, limited 1, stained 4
tier	1	86	0.776	0.0%	clear 58, limited 4, normal 17, stained 7
tier	3	17	0.387	0.0%	clear 14, stained 1, normal 2
strict	same-lake	86	0.689	1.2%	clear 65, limited 4, normal 12, stained 5
strict	strict	22	0.746	0.0%	clear 9, stained 4, limited 2, normal 7
wisconsin	WI	95	0.743	0.0%	clear 63, normal 19, limited 4, stained 9
wisconsin	outside WI	13	0.661	7.7%	clear 11, limited 2
season	cool	10	0.633	10.0%	clear 5, limited 2, stained 2, normal 1
season	warm	98	0.732	0.0%	clear 69, normal 18, limited 4, stained 7

- **Where the gate helps.** On the same rows, the Stage 15 semantics gave 4.6% harmful. The gate gives 0.9%, the single harmful answer being a tier-2 Normal-to-Stained on 1.6 ft water.
- **Why it is not a clean test.** 53% of the rows are Sentinel-2C, at lower authority, so they can never be harmful; and 88% are Wisconsin, mostly clear water.
- **The strict lake-independent subset** (22 rows) scores 0.746 with no harmful answer.

6. Exact feet

Exact feet (Stage 12 rubric)	Fresh: published (coverage), signed	Fresh verdict	Development verdict (Stage 16 semantics)
Clear	34 (46%), -2.1 ft	NOT	PARTIAL
Normal	3 (16%), +0.3 ft	NOT	SUPPORTED
Stained	5 (56%), +0.8 ft	PARTIAL	PARTIAL
Limited	5 (83%), +0.4 ft	SUPPORTED	SUPPORTED
Algae	–	NOT	SUPPORTED
Sediment	–	NOT	SUPPORTED
Bottom-risk	–	NOT	NOT
Score		**2/10**	**7/10**

- **Feet accuracy is not the problem.** The fresh 2/10 comes from coverage. Exact feet are withheld on Sentinel-2C (53% of fresh rows), and three classes are short. Where feet are shown (non-S2C), the error is ×1.32 with a median −0.6 ft.
- **On development,** under the same semantics, exact feet score 7/10. That is one point lower than Stage 14, because Unknown now withholds feet.

7. Sector Intelligence suitability

Sector Intelligence evidence	Rule	Development share / truth in answer	Fresh n / share / truth in answer	Bar	Fresh verdict
Strong	full authority, tier 1, not S2C, no algal flag, posterior ≥ 0.90	10% / 97%	5 / 5% / 100%	0.90	INSUFFICIENT (n < 30)
Moderate	other full-authority answers	70% / 85%	39 / 36% / 97%	0.80	met
Weak	lower authority (tier 3, S2C)	7% / 85%	55 / 51% / 85%	0.65	met
Unavailable	Unknown	13% / –	9 / 8% / –	–	–

SAFE FOR SECTOR INTELLIGENCE: NO, by the pre-registered rule: the strong level has only 5 fresh answers, against 30 needed. On the rows available:

- moderate evidence was right 97% of the time;
- weak evidence was right 85% of the time.

Development shows the shape a feed would have: strong 10% (right 97%), moderate 70% (85%), weak 7% (85%), unavailable 13%.

8. Ready to freeze?

Criterion (unchanged from Stage 15)	Development	Fresh	Bar
Core minima met	n/a	**no** (normal, stained, limited, total)	all
Operational	7/10	7/10	≥ 7/10
Every core class ≥ 0.60	**no** (stained 0.571)	(normal 0.529; short classes insufficient)	≥ 0.60
Harmful	10.6%	0.9%	≤ 15%
Severe	1.5%	0.0%	≤ 2%

READY TO FREEZE CLARITY AND START WATER TEMPERATURE: NO.

- **What it now meets.** On development the gate meets the harmful and severe bars for the first time: 10.6% and 1.5%.
- **What still fails:**
  - stained water stays at 0.571;
  - the fresh holdout cannot certify anything until it is about twice as large overall, and several times larger in stained and limited water.

9. Stretch target: a true 9/10 product

Stretch target (true 9/10)	Development	Fresh	Target
Operational	0.693	0.709	≥ 0.85
Weakest class	0.571	0.529	≥ 0.80
Harmful	10.6%	0.9%	≤ 5%
Severe	1.5%	0.0%	≤ 0.5%
Strong evidence	10%	5%	≥ 50%

10. The shortage, and the sources that would complete it

The size of the gap.

- **The pool.** All 2026 WQP data at untouched sites gives 1,111 events, but only 108 matched observations. Of those, 9 are stained and 6 limited, against minima of 60 and 25.
- **What would close it.** At this yield (about 10% of events match, and 8% of matches are stained), 60 stained rows need about 7,500 eligible events, roughly 7× today's pool, in murkier water than Wisconsin's.

Independent sources that would close it, all before anyone looks at the values:

1. **The 2026 agency data not yet in WQP.** Many states upload a season's data the following spring: FL LAKEWATCH, MN PCA, TCEQ, ADEM, Iowa DNR. A re-pull in about April 2027, under this same frozen protocol, is the cheapest route.
2. **Southern reservoir agencies, by direct request.** TVA Reservoir Ecological Health monitoring, and USACE district water-quality programs (Nashville, Mobile, Vicksburg, Tulsa, Fort Worth). This is where stained and limited water lives.
3. **State reservoir programs with their own portals:** Oklahoma OWRB BUMP, Missouri LMVP, Texas SWQMIS, Iowa AQuIA, Illinois VLMP.

11. Path from the current score to a true 9/10

The arithmetic. A product whose every answer is a correct boundary state scores 0.90, so 9/10 (≥ 0.85 class-balanced) needs about 90% of every class answered correctly at full authority. Today that is 0.69–0.71. Four things stand in the way:

1. **Resolution.** Stained water (2–3 ft) is narrower than the retrieval error (×1.36), so stained answers are mostly boundary states or Unknown (0.57 development). A 9/10 needs more information per answer than one scene can give, such as multi-scene consensus or a later design that assimilates field reports. That means reopening model work with a new pre-registration.
2. **Authority.** Lower-authority answers cap at 0.75 and Unknown at 0.5. Sentinel-2C is a third of current scenes (53% of 2026 matches), and it needs to pass its lift criterion: more tandem pairs, and S2C harmonisation re-derived

against the low bias now seen on both independent sets.

3. **Evidence.** A fresh holdout of ≥ 200 rows with ≥ 60 stained (section 10) is needed to certify any of this.
4. **Strong evidence** is 5–10% of answers. Sector Intelligence would need most answers at that level.

Files

Everything is in `scratchpad/stage16/`:

Path	Contents
`PREREGISTRATION_STAGE16.md`, `FREEZE_STAGE16.json`	The frozen plan and every hash, in order
`engine/engine16.py`, `engine/rule16.json`	The gate and tier-2 rule over the unchanged engine
`analysis/lib16.py`, `dev16.py`, `dev16.json`	The posterior gate and its development derivation
`fresh/`	The fresh WQP pull (`wqp/`), value-blind index, prescreen, matches, `fresh_sample.csv`, ACOLITE results, `truth16.json`
`analysis/score16.py`, `score16.json`	The one-time fresh score

FROZEN ENGINE CHANGED: NO

SINGLE-STATE RELIABILITY GATE: Posterior gate. A single state needs a ≥ 70% calibrated posterior and a boundary pair ≥ 70%; otherwise Unknown. It was fitted on development only. It never issues plain Normal or Stained (zones: Limited < 1.55 ft, Stained-to-Limited, Unknown 2.43–2.80 ft, Normal-to-Stained, Clear-to-Normal 4.11–7.40 ft, Clear above). Development harmful 16.6% → 10.6%, severe 2.0% → 1.5%

TIER-2 AUTHORITY RULE: Boundary states only, with tier-2-calibrated posteriors; Unknown below 70%. Tier-2 harmful on development 12% → 7% (the Stage 15 re-score had shown 29%)

FRESH HOLDOUT SAMPLE SIZE: 108 observations at 80 lakes (88% Wisconsin, 57 Sentinel-2C). INSUFFICIENT: normal 19/40, stained 9/60, limited 6/25, total 108/200 (clear 74/40 met)

HARMFUL RATE: 0.9% on the fresh holdout (1 of 108; insufficient, and S2C rows are lower authority so cannot be harmful); 10.6% on development

SEVERE RATE: 0.0% on the fresh holdout; 1.5% on development

CLEAR SCORE: 0.826 fresh (n 74); 0.880 development

NORMAL SCORE: 0.529 fresh (n 19, insufficient); 0.643 development

STAINED SCORE: 0.706 fresh (n 9, insufficient); 0.571 development (below 0.60)

LIMITED SCORE: 0.775 fresh (n 6, insufficient); 0.676 development

EXACT-FEET CAPABILITY: 2/10 on the fresh holdout (S2C feet withheld on 53% of rows; three classes insufficient; where shown, ×1.32 and −0.6 ft); 7/10 on development under Stage 16 semantics

OPERATIONAL BOWFISHING CLARITY: 7/10 on the fresh holdout (0.709; insufficient for normal, stained and limited); 7/10 on development (0.693)

SAFE FOR SECTOR INTELLIGENCE: NO (only 5 strong-evidence answers against the 30 needed; moderate evidence 97% and weak 85% right on the rows available)

READY TO FREEZE CLARITY AND START WATER TEMPERATURE: NO (fresh minima not met; development stained 0.571 < 0.60; harmful and severe now pass on development)

PATH FROM CURRENT SCORE TO TRUE 9/10: (1) Certify first: a fresh holdout of ≥ 200 rows with ≥ 60 stained and ≥ 25 limited, from a WQP re-pull around April 2027 (2026 uploads) or direct TVA/USACE and state reservoir requests,

under this frozen protocol. (2) Resolution: stained water is narrower than the ×1.36 retrieval error, so a 9/10 needs more information per answer (multi-scene consensus or field-report assimilation), which means reopening model work with a new pre-registration. (3) Authority: Sentinel-2C (a third of current scenes) must pass its lift criterion; it still reads low on both independent sets (−0.19 and −0.09 log), so it needs re-derived harmonisation from more tandem pairs. (4) Raise strong evidence from 5–10% toward half of all answers. Today: 0.69–0.71 class-balanced, against the 0.85 a 9/10 needs